KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2018/2019

SECOND SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE (EE, AEROSP,BIOS,BIOM,PET,MECH AND CIVIL ENGINEERING)

ECU 103: PHYSICS FOR ENGINEERS II

ATE: TUESDAY, 9TH APRIL, 2019 **TIME: 2.-00 P.M. - 4.00 P.M.**

NSTRUCTIONS:

1. Answer questions ONE and any other TWO questions.
2. The following constants and table may be useful:
   $c = 3 \times 10^8$ m/s, $\varepsilon_o = 8{:}85 \times 10^{-12}$ F/m, $\mu_o = 4\pi \times 10^{-7}$ H/m
   $\sin A + \sin B = 2\sin[(A + B)/2]\cos[(A-B)/2]$

Q1. (a) (i) In words, define superposition principle. (1 mark)

(ii) Three charges of magnitude 2 μC are placed on the vertices of an equilateral triangle of side 100 mm. Calculate the resultant force on one charge due to the other charges (3 marks)

(b) (i) Give the mathematical formula of electric potential in terms of scalar dot product. (1 mark)

(ii) A 4 μC charge is located at (0, 2) m and a -2 μC charge is located at (0, -2) m. Find the potential at (0, -1) m. (3 marks)

(c) (i) In words, define of Gauss's law in electrostatics? (1 mark)

(ii) What is the capacitance of a 1 m long cylindrical capacitor with inner and outer conductors radii of 1 mm and 1.5 mm, respectively? (3 marks)

(d) (i) In words, define of Ampere's law. (1 mark)

(ii) A long straight wire of radius 1.5 mm carries a current of 32 A. What is the magnetic field at 1.2 mm from the axis. (3 marks)

(e) (i) In words, what is the meaning of dielectric constant of material? (1 mark)

(ii) A parallel-plate capacitor whose capacitance is 13.5 pF is connected to a 12.5 V battery. The battery is disconnected and a dielectric slab with relative permittivity 6.5 is inserted between the plates. What is the potential energy after the slab is introduced (3 marks)

(f) (i) Give one function of the objective lens of a telescope. (1 mark)

(ii) Locate the image when a point object in air falls on a convex refracting surface 20 cm away with a radius of curvature of 10 cm before entering a medium whose refractive index is 2. (3 marks)

(g) (i) In words, explain the meaning of work function of a metal? (1 mark)

(ii) What is the energy of photons in a beam of UV light of wavelength 240 nm? (3 marks)

(h) Give three characteristics of an image formed by a convex mirror. (2 marks)

Q2. (a) A beam of X-rays of wavelength 0.24 nm is directed towards a sample. The X-rays scatter at various angles from the electrons within the sample, imparting momentum to the electrons, which are initially at rest.

(i) What is the longest wavelength measured by a detector? (3 marks)

(ii) At what scattering angle does this occur? (3 marks)

(iii) For this scattering angle, what is the kinetic energy of the recoiling electrons? (3 marks)

(iv) If the detector measures a wavelength for the scattered X-rays of 0.2412 nm, what is the X-rays scattering angle? (3 marks)

(b) show that ~~Find~~ the electric field at a point P above the midpoint of a straight line segment of length 2L which carries a uniform line charge λ as shown in fig. 1.

is given by $E = \dfrac{L\lambda}{2\pi\varepsilon_0 z\sqrt{z^2+L^2}}$

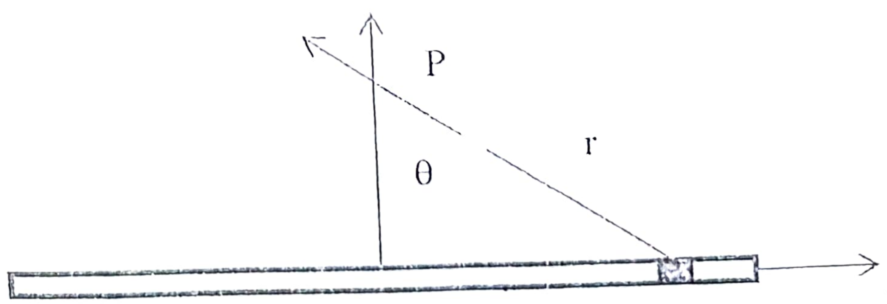


Fig. 1

(8 marks)

KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2014/2015

SECOND SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE (ELECTRICAL AND ELECTRONIC ENGINEERING)

ECU 103: PHYSICS FOR ENGINEERS II

DATE: Thursday, 16th April 2015 TIME: 11.00 a.m. – 1.00 p.m.

INSTRUCTIONS:

1. Answer questions ONE and any other TWO questions.
2. The following constants and table may be useful:

   $c = 3 \times 10^8$ m/s, $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m, $\mu_0 = 4\pi \times 10^{-7}$ H/m

Q1.(a) (i) In words, explain the meaning of superposition principle in electrostatics? (1 mark)

(ii) Three identical charges are placed on the vertices of an equilateral triangle. If two charges exert a force of 0.001 N each on the third. What is the resultant force on the third charge? (3 marks)

(b) (i) In words, explain the meaning of Biot-Savart law? (1 mark)

(ii) A curved wire subtending an angle $\pi/4$ at the center of curvature of radius 5 cm, is carrying 5 A. Find the field at the center of curvature. (3 marks)

(c) (i) In words, explain the meaning of Gauss's law? (1 mark)

It states that the electric flux thru a hypothetical surface. is equal to ... enclosed

(ii) Use Gauss's law to derive the expression for capacitance per unit length of a cylindrical capacitor. (3 marks)

(d) (i) In words, explain the meaning of Amperes law? (1 mark)

(ii) A wire of radius 3 mm carrying 2 A. Find the magnetic field at a point 2 mm from its center (3 marks)

(e) (i) In words, explain the meaning of a filter? (1 mark)

ii) Show that an RC circuit can act as a low pass filter. (3 marks)

(f) (i) Give one application of a convex mirror? (1 mark)

(ii) A convex lens has radii of curvature of 40 cm is made of glass with $n = 1.65$. Find its focal length. (3 marks)

(g) (i) In words, explain the meaning of Huygen's principle? (1 mark)

(ii) A light beam passes from medium 1, with an index of refraction n_1, through a thick slab of material whose index of refraction is n_2, and emerges from the slab into medium 1. Show that the emerging beam is parallel to the incident beam. (3 marks)

(h) Differentiate between linearly polarized and circularly polarized waves. (2 marks)

Q2. (a) A parallel plate capacitor of area 115 cm^2 and plate separation 1.24 cm is filled with removable dielectric slab with $\varepsilon_r = 2.6$. The capacitor is given a charge Q with the slab removed and not disconnected from the battery of $V_o = 90$ V. Find

(i) the charge on the plates without the slab. (2 marks)

(ii) the charge on the plates with the slab inserted. (2 marks)

(iii) the field without the slab. (2 marks)

(iv) the field with the slab inserted. (2 marks)

(v) induced charge with the slab inserted (2 marks)

(b) A 2 μF capacitor is charged using a 100 V battery. The battery is disconnected and two uncharged capacitors are connected to it as shown in Fig. 1.

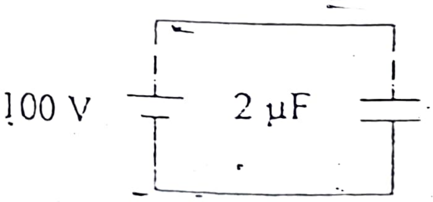


Fig. 1a

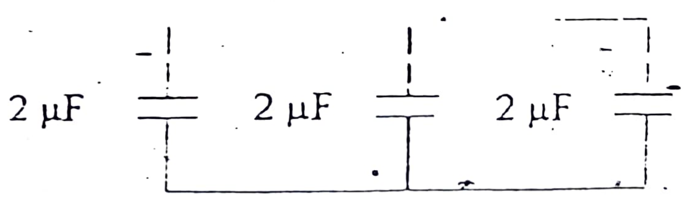


Fig. 1b

Find (i) the energy stored in Fig. 1a (5 marks)

(ii) the energy stored in Fig. 1b (5 marks)

Q3. (a) Consider the circuit in Fig 2 below carrying 2 A. Find the magnetic field at point P.

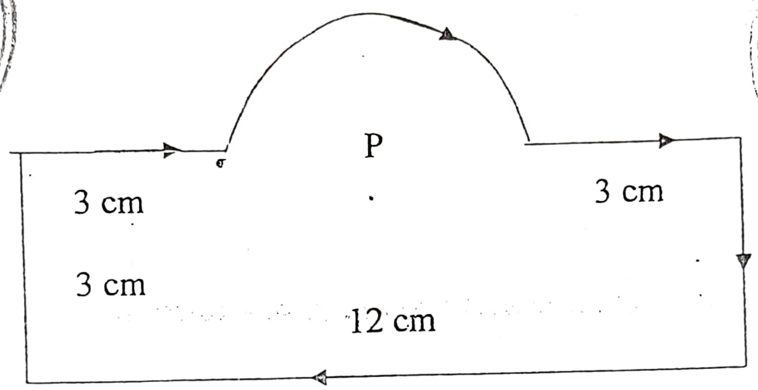


Fig. 2

$R = 250\,\Omega$

(b) In an RLC circuit for which L = 0.6 H, C = 3.5 μF, ω = 377 rad/s and V_m = 150 V, find the

(i) impedance (2 marks)

(ii) maximum current (2 marks)

(iii) phase angle (2 marks)

(iv) voltage across resistor, capacitor and inductor (2 marks)

(v) average power delivered (2 marks)

Q4. (a) Show that the intensity pattern of a double-slit diffraction experiment is given by

$$I_\theta = I_m(\cos\beta)^2\left(\frac{\sin\alpha}{\alpha}\right)^2$$

where $\beta = \dfrac{\pi d\sin\theta}{\lambda}$ and $\alpha = \dfrac{\pi a\sin\theta}{\lambda}$ and a and d are slit width and separation, respectively. (8 marks)

(b) In double-slit Fraunhofer diffraction, what is the

(i) fringe spacing on a screen 50 cm away from the slits if they are illuminated with the blue light ($\lambda = 4800$ A), if $d = 0.1$ mm, and if $a = 0.02$ mm? (4 marks)

(ii) linear distance from the central maxima to the first minima on the fringe envelope? (4 marks)

(iii) number of fringes in the central peak? (4 marks)

5. (a) Light of wavelength 500 nm is incident normally on a diffraction grating. If the third order maximum of the diffraction pattern is observed at an angle of 32°

(i) what is the number of rulings per cm for the grating? (3 marks)

(ii) Determine the total number of primary maxima that can be observed in this situation. (3 marks)

(b) For a convex refracting surface of radius of curvature r, show the object and image distances are related by the equation

$$\frac{n_1}{o} + \frac{n_2}{i} = \frac{(n_2 - n_1)}{r}$$

(8 marks)

(c) In a Newton's rings experiment the radius of curvature of the lens is 5 m and its diameter is 2 cm.

(a) How many rings are produced? (3 marks)

(b) How many would rings be seen if the arrangement were immersed in water (n = 1.33, $\lambda = 5890$ A) (3 marks)

BACHELOR OF SCIENCE (MECHANICAL ENGINEERING),
BACHELOR OF SCIENCE (CIVIL ENGINEERING),
BACHELOR OF SCIENCE(PETROLEUM ENGINEERING), AND
BACHELOR OF SCIENCE (BIOMEDICAL & BIOSYSTEMS ENGINEERI

ECU 103: PHYSICS FOR ENGINEERS II

DATE: Wednesday, 22nd August 2018 **TIME:** 8.00 a.m. - 10.00 a.m.

NSTRUCTIONS

Time constant

1. Answer questions ONE and any other TWO questions.
2. The following constants and table may be useful:
   $c = 3 \times 10^8$ m/s, $\varepsilon_o = 8.85 \times 10^{-12}$ F/m, $\mu_o = 4\pi \times 10^{-7}$ H/m
   $\sin A + \sin B = 2\sin[(A + B)/2]\cos[(A-B)/2]$

Q1. (a) (i) In words, define Coulomb's law. (1 mark)

(ii) Point charges 1 mC and -2 mC are located at (3, 2, -1) and (-1, -1, 4), respectively. Calculate the electric force on a 10 nC charge located at (0, 3, 1)? (3 marks)

(b) (i) In words, define electric potential at a point? (1 mark)

(ii) Two point charges – 4 μC and 5 μC are located at (2, -1, 3) and (0, 4, -2), respectively. Find the potential at (1, 0, 1). (3 marks)

(c) (i) In words, define of Gauss's law in electrostatics? (1 mark)

(ii) What is the capacitance of a spherical capacitor with inner and outer radii of 1 mm and 1.5 mm, respectively? (3 marks)

(d) (i) In words, define of Ampere's law. (1 mark)

(ii) A long straight wire of radius 1.5 mm carries a current of 32 A. What is the magnetic field at 1.2 mm from the axis. (3 marks)

(e) (i) In words, what is meaning of a dielectric material? (1 mark)

(ii) A parallel-plate capacitor whose capacitance is 13.5 pF is connected to a 12.5 V battery. The battery is disconnected and a dielectric slab with relative permittivity 6.5 is inserted between the plates. What is the potential energy after the slab is introduced (3 marks)

(f) (i) Give one function of the eye piece of a telescope. (1 mark)

(ii) Locate the image when a point object in air falls on a convex refracting surface 20 cm away with a radius if curvature of 10 cm before entering a medium whose refractive index is 2. (3 marks)

(g) (i) In words, explain the meaning of diffraction of light? (1 mark)

(ii) Coherent light of wavelength 580 nm is incident on a slit of width 0.3 mm. If the observing screen is placed 2 m from the slit, find the position of the first dark fringe. (3 marks)

(h) Differentiate between Kirchhoff's rules for current and voltage. (2 marks)

Q2. (a) Show that for double-slit diffraction, the intensity on the screen is given by.

$$I_\theta = I_m(\cos\beta)^2\left(\frac{\sin\alpha}{\alpha}\right)^2$$

where I_m, β and α are $(2E_o)^2$, $(\pi d\sin\theta)/\lambda$ and $\beta = (\pi a\sin\theta)/\lambda$. a and d slit width and slit separation respectively. (8 marks)

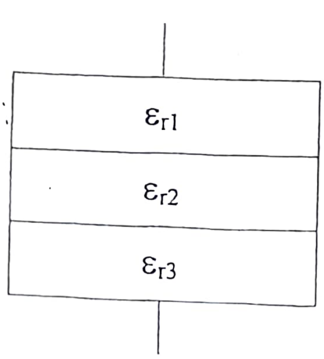
(b) In double-slit Frauhofer diffraction, what is the

(i) fringe spacing on a screen 50 cm away from the slits if they are illuminated with blue light ($\lambda = 4800$ A, d = 0.1 mm and a = 0.02 mm) (4 marks)

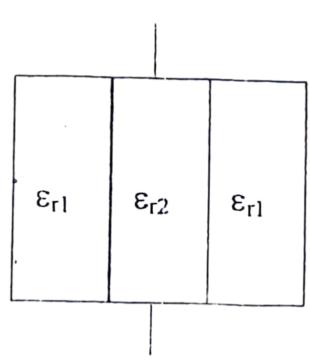
(ii) linear distance from the central maxima to the first minima on the fringe envelope. (4 marks)

(iii) the number of fringes in the central peak. (4 marks)

Q3. (a) Determine the capacitance of each of the capacitors in Fig. 1a and b.



(a)



(b)

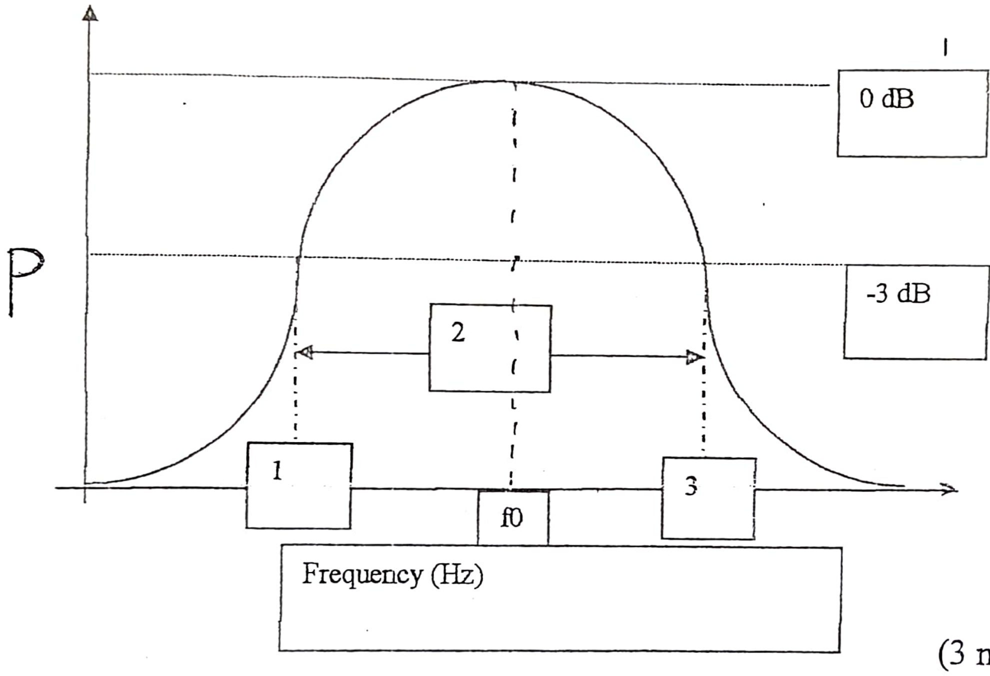
(14 marks)

(b) A cylindrical capacitor has radii 1 cm and 2.5 cm. If the space between the plates is filled with a dielectric with $\varepsilon_r = (10 + \rho)/\rho$, find the capacitance per meter. (6 marks)

4a. i). What is the role of a band-pass filter in electric circuits? (1 mark)

ii). Describe briefly the following terms with respect to band-pass filters; 1. Pass band and 2. Q-factor. (4 marks)

b). The graph drawn below shows the magnitude transfer function versus the frequency for a band-pass filter output in a simple circuit. Name the sections marked as 1), 2) and 3)



(3 marks)

c). State and briefly explain a specific area of application in engineering where band-pass filters are used. (4 marks)

d). Sketch three different types of lenses represented by the two group of lenses below;

i). Convergent group of lenses. (1 mark)

ii). divergent group of lenses. (1 mark)

e). i). With the help of the ray-diagram provided below, derive the optical expression;

$$\frac{1}{u} - \frac{1}{v} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

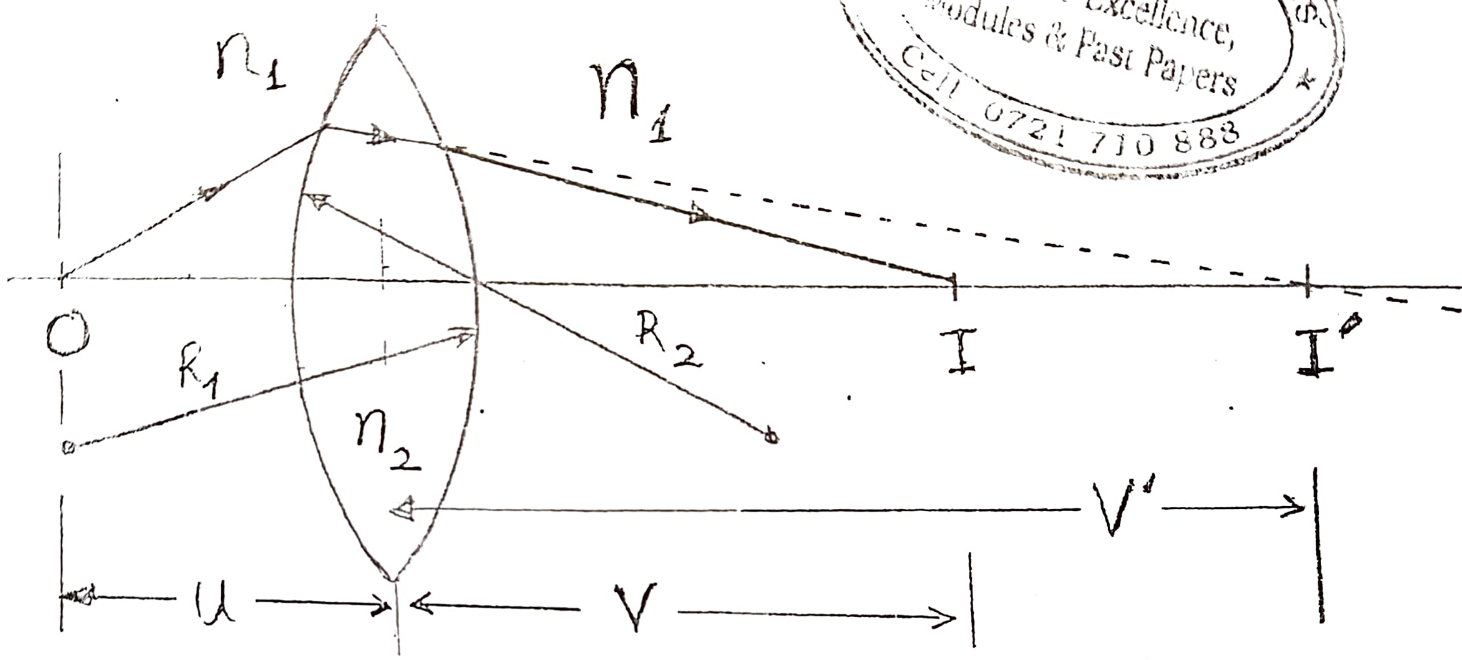
Where; n is the refractive index of the lens,

R_1 and R_2 are the radii of curvature of the two co-axial spherical surfaces of the lens and

u and v are respectively object and image distances. (4 marks)

Mention two conditions under which the equation (in 4ei.) above is valid. (2 marks)

Ray-diagram for Q4 ei.



5a. i). Explain the phenomenon of natural radioactivity in some elements. (2 marks)

ii). How can artificial radioactivity be induced in some elements? (2 marks)

iii). The half life of carbon-14 isotope is 5577 years. Calculate the time in years required for the isotope's activity to reduce to a fifth of its original. (4 marks)

b. i). With the help of a flow chart diagram, explain clearly the steps involved in the production of nuclear energy from a reactor. (6 marks)

ii). Mention one significant safety precaution to be observed while handling radioactive materials. (1 mark)

c. i). Sketch a setup of an arrangement capable of producing X-rays. (4 marks)

ii). Mention one clear advantage of using X-rays in a professional area of your choice. (1 mark)

BACHELOR OF SCIENCE (AEROSPACE ENGINEERING),
BACHELOR OF SCIENCE-(BIOMEDICAL ENGINEERING),
BACHELOR OF SCIENCE, (PETROLEUM ENGINEERING),
BACHELOR OF SCIENCE (MECHANICAL ENGINEERING AND
BACHELOR OF SCIENCE (CIVIL ENGINEERING)

ECU 103: PHYSICS FOR ENGINEERS II

TE: Friday, 16th June 2017 TIME: 8.00 a.m. - 10.00 a.m.

INSTRUCTIONS

1. Answer questions ONE and any other TWO questions.
2. Question one carries 30 marks while the rest carry 20 marks each.
3. The following constants and information may be useful:
   $c = 3 \times 10^8$ m/s, $\varepsilon_o = 8.85 \times 10^{-12}$ F/m, $\mu_o = 4\pi \times 10^{-7}$ H/m.
   $1/cosec\theta = sin\theta$ $\quad\quad cot^2\theta + 1 = cosec^2\theta$

QUESTION ONE

a). i). Define in words, the electric displacement at a point? (1 mark)
ii). What is the electrostatic electric displacement at a point with an electric field of 1000 Vm^{-1}? (3 marks)

b). i). Define in words, the electric field at a point. (1 mark)
ii). What is the electric field at a distance of 2 m from a 300μC charge? (3 marks)

c). i). Give a statement of Gauss's electrostatics law. (1 mark)
ii). Use Gauss's law to derive the expression for a cylindrical-plate capacitor (C) of surface area (A), with inner and outer radii dimensions a and b respectively. (3 marks)

d). i). Give a mathematical equation for the Biot-Savart law. (1 mark)
ii). Two parallel wires 1 and 2, carrying opposite currents of magnitude 2A and 3A respectively, are 2 cm apart and 10 cm long each. What is the magnetic force on wire2 due to wire*1*. (3 marks)

e). i). Define the term, time constant of an RL series circuit. (1 mark).
ii). Find the angular frequency of oscillation in an LC circuit with inductance and capacitance values of 3 mH and 3 μC respectively. (3 marks)

i). Give one application of a convex mirror in engineering? (1 mark)

ii). Draw a ray-diagram illustrating the functioning of a basic telescope. (3 marks)

g). i). Briefly explain the phenomenon of light diffraction. (1 mark)

ii). Coherent light of wavelength 580 nm is incident on a slit of width 0.3 mm. If the observing screen is placed 2 m from the slit, find the position of the first dark fringe. (3 marks)

h). Differentiate between Kirchhoff's current and voltage rules. (2 marks)

QUESTION TWO

(a) In the circuit shown in Fig. 1, find currents I_1 , I_2 and I_3. (9 marks)

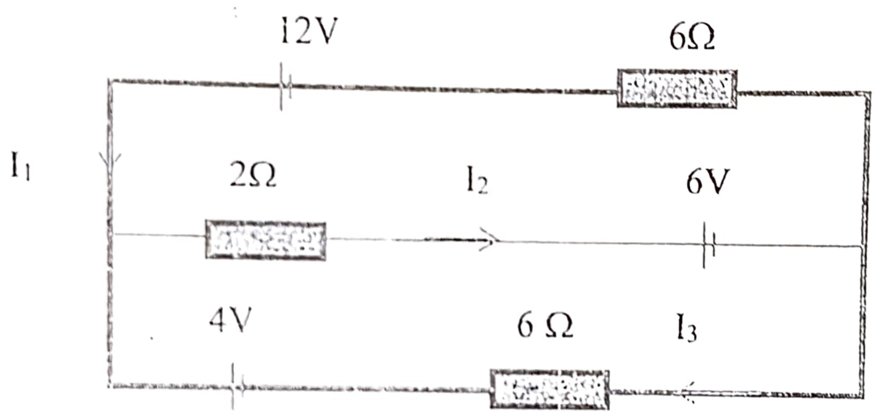


Fig. 1

(b) A voltmeter of resistance 500 Ω, placed in Fig.2 reads 50 V, determine the following;

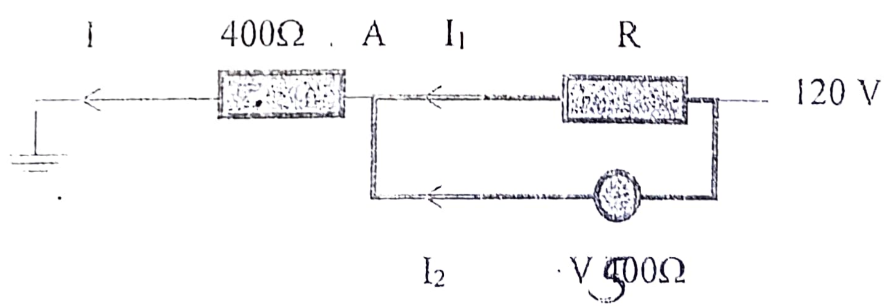


Fig. 2

i). Resistance R. (3 marks)

ii). The potential at point A when the voltmeter is not in the circuit. (2 marks)

iii). Currents I, I_1 and I_2. (6 marks)

QUESTION THREE

a). Consider the circuit in Fig 3 below carrying a 2A current as indicated by the arrow. Find the magnetic field at point P. (11 Marks)

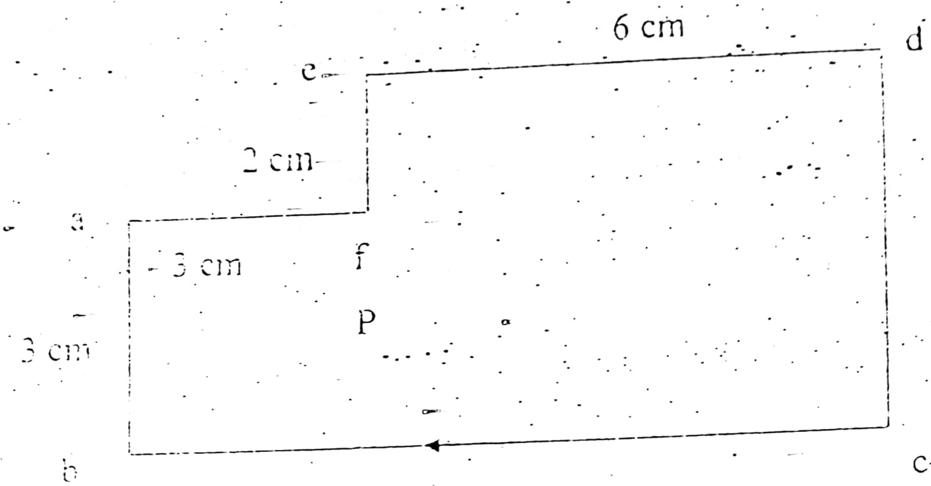


Fig. 3

b). A square loop of wire with sides 5 cm, falls at a speed of v under influence of gravity past a region with uniform magnetic field of 15 T, then into a region with no magnetic field as shown in fig. 4.
The total resistance of the loop is 1 Ω and its mass is 150g. Find the following;

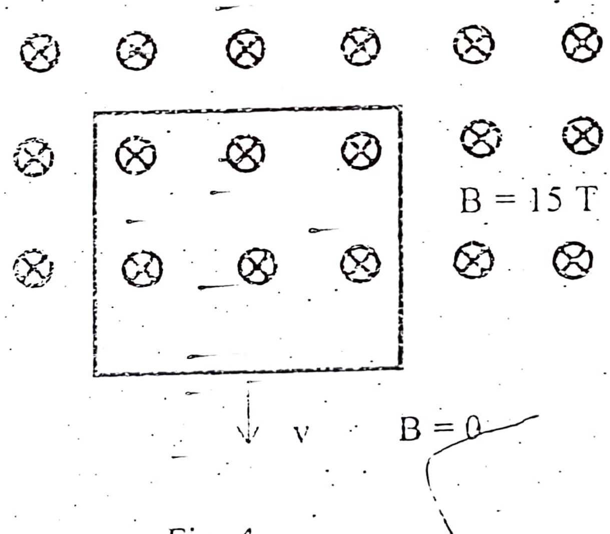


Fig. 4

i). The terminal velocity of the loop as it passes the boundary between the two fields. (3 marks)

ii). Power dissipated in the loop. (3 marks)

iii) Total energy lost due to joule heating in the loop. (3 marks)

QUESSTION FOUR

a). Two flat glass plates of length 10 cm each, touch at one end but are separated by a thin wire of diameter $d = 0.01$ mm at the other end as shown in fig. 5 below. What is the;

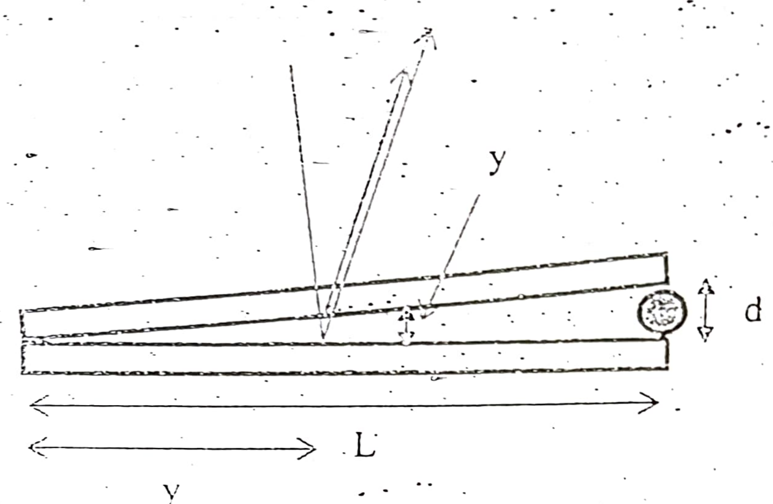


Fig. 5

i). Distance x between the observed maxima, if the incident light has a wavelength 420 mm? (5 marks)

ii). Total number of bands of constructive interference. (5 marks)

(b) A soap bubble of thickness t units has a refractive index n. Light of wavelength λ in air falls vertically on the bubble and is reflected back as shown in Fig. 6. Below.

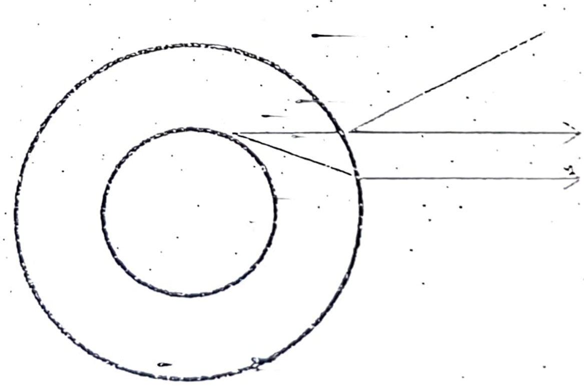


Fig. 6

i). Express the condition for constructive interference for the reflected light. (5 marks)

ii). If t is 400 nm and n = 1.3, what wavelengths will interfere constructively in the reflected light? (5 marks)
(Assume m = 1, 2, 3 and 4).

(a) Consider the circuit in Fig 2 below carrying 2 A. Find the magnetic field at point P.

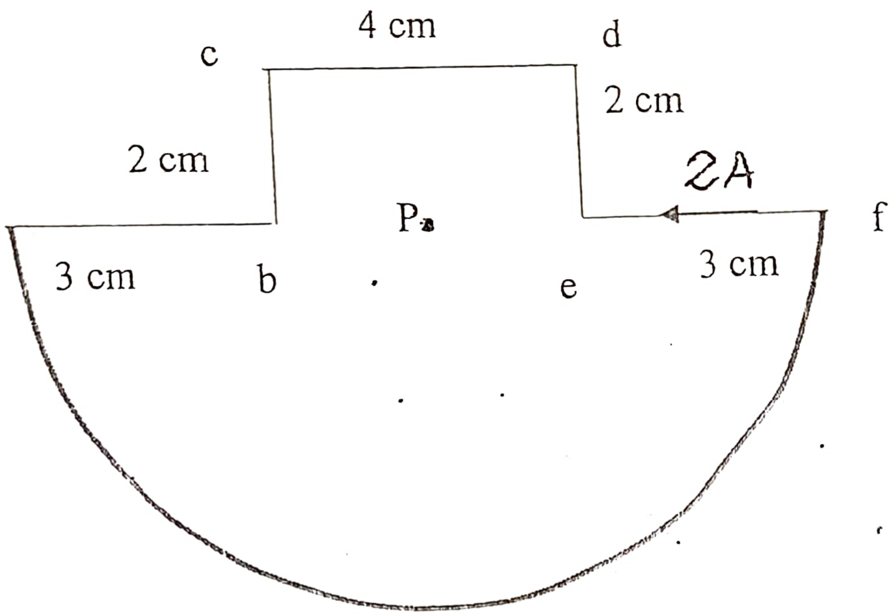


Fig. 2.

(12 marks)

(b) An infinitely long wire carrying a current I is shown in fig. 3. Show that ~~Find~~ the expression for magnetic field at point P. is given by $B = \frac{\mu_0 i}{2\pi r}$

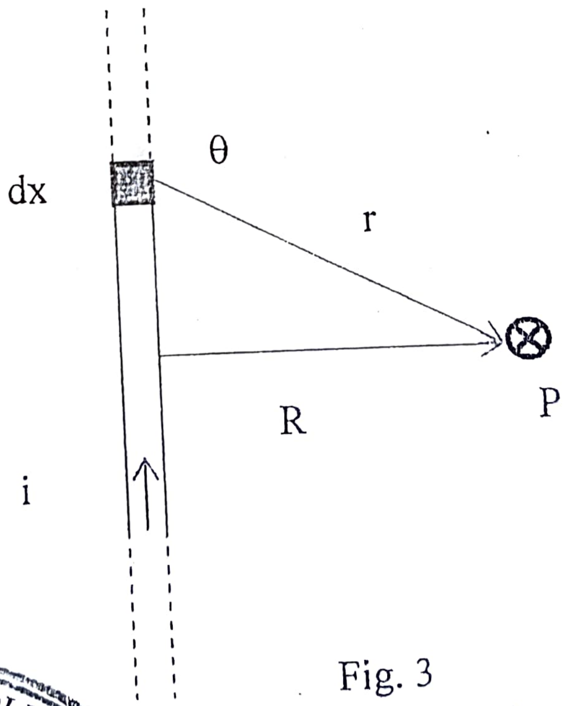


Fig. 3

(8 marks)

Q4. (a) Consider a circuit consisting of a conducting rod of length l sliding ~~on~~ along two fixed parallel conducting rails as shown in fig. 4.

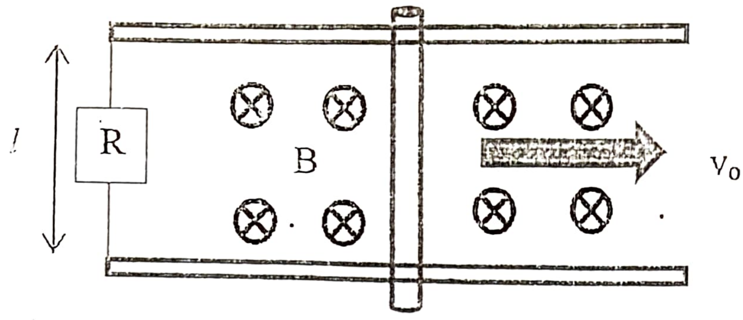


Fig. 4

If the rod is given an initial velocity v_o to the right and released show that the

(i) velocity of the bar is given by

$$v = v_o exp\left[-\frac{t}{\tau}\right]$$ (6 marks)

(ii) current in the bar is given by

$$I = \frac{Blv_o}{R} exp\left[-\frac{t}{\tau}\right]$$ (2 marks)

(iii) induced e.m.f. in the circuit is given by

$$v = Blv_o exp\left[-\frac{t}{\tau}\right]$$ (2 marks)

where $\tau = mR/(B^2 l^2)$

(b) A square loop of wire with sides 5 cm falls at a speed v under the influence of gravity past a region with uniform magnetic field of 15 T into a region with no magnetic field as shown in Fig. 5.

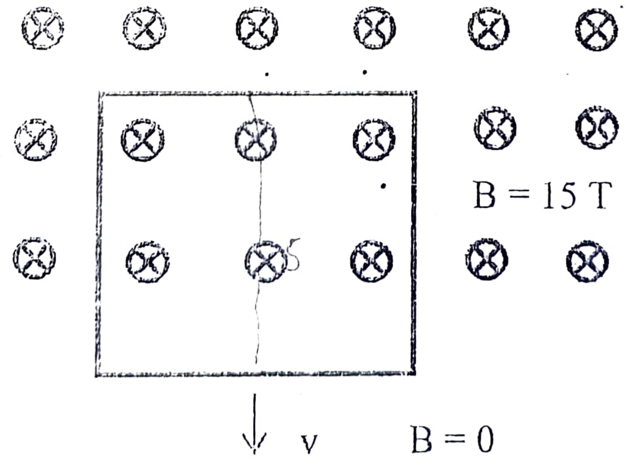


Fig. 5

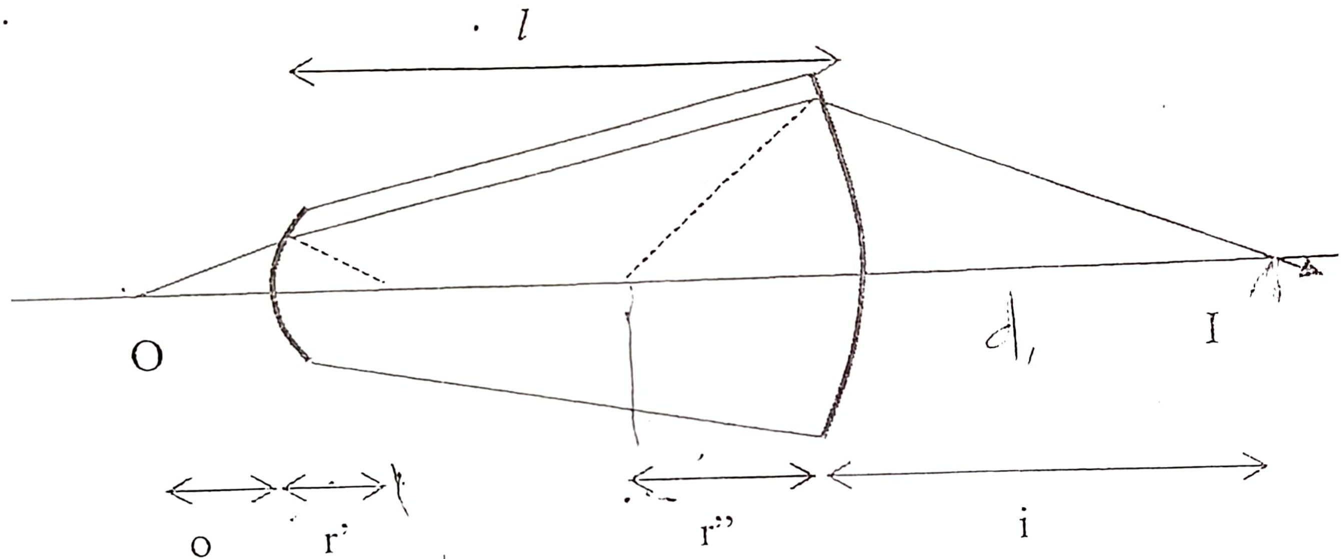
The total resistance of the loop is 1 Ω and its mass is 150 g. Find the

(i) terminal velocity of the loop as it passes the boundary between the two fields. (4 marks)

(ii) power dissipated in the loop. (3 marks)

(iii) total energy lost to joule heating in the loop. (3 marks)

(a) Fig. 6 shows a glass lens of thickness l and of refractive index n whose surface are found to radii r' and r'' respectively.



d_o

Fig. 6

For lens approximation, show that the focal legthth is given by

$\frac{1}{f} = (n-1)\left(\frac{1}{r'} - \frac{1}{r''}\right)$ (10 marks)

b) A biconvex lens has radii of curvature of 20 cm. The lens is fitted into the side of a water tank. If a point object is place a distance o from the lens along the axis,

(i) find the expression for the image position (i'') in terms of n_g, n_w, n_a r'', r' and o. (7 marks)

(ii) If $n_g = 1.5$, o = 100 cm, r'' = -20 cm, r' = 20 cm and $n_w = 1.34$, calculate i''. $n_a = 1$ (3 marks)

$h = 6.6 \times 10^{-34}$

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D SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE

(EE.AEROSP. BIOS. BIOM. PET. MECH. AND CIVIL ENGINEERING)

ECU 103 PHYSICS FOR ENGINEERS II

WEDNESDAY 13TH APRIL 2016 TIME: 8.00 A.M. - 10.00 A.M.

UCTIONS

Answer questions ONE and any other TWO questions.

The following constants and table may be useful:

$c = 3 \times 10^{8}$ m/s, $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m, $\mu_0 = 4\pi \times 10^{-7}$ H/m

$1/\operatorname{cosec}\theta = \sin\theta A \qquad \cot^2\theta + 1 = \operatorname{cosec}^2\theta$

(a) (i) In words, define electrostatic potential at a point? (1 mark)

(ii) A 2 μC charge is located at the origin. What is the electrostatic potential at (0, 2) m? $Q = CV$, $V = \frac{Q}{4\pi\varepsilon_0 r}$ (3 marks)

(b) (i) In words, define Coulomb's law? (1 mark)

$Q = It$

(ii) A $Q_1 = 300$ μC charge is located at (1, -1, -3) m experiences an $8\hat{x} - 8\hat{y} + 4\hat{z}$N force due to a charge Q_2 located at (3, -3, -2) m. Find Q_2. $\frac{Q_1 Q_2}{4\pi\varepsilon r^2}$ (3 marks)

(c) (i) In words, define of Gauss's law in electrostatics? (1 mark)

$Q = \varepsilon EA$

(ii) Use Gauss's law to derive the expression for a parallel-plate capacitor of area A and plate separation d. $Q = CV$, $C = \frac{\varepsilon_0 A}{d}$ (3 marks)

(d) (i) In words, define Amperes law? (1 mark)

(ii) Wires 1 and 2 carrying currents 2 and 3 A, respectively, are in parallel, 2 cm apart and each 10 cm long. Find the magnetic force on wire 2 due to wire 1. (3 marks)

$B = \mu_0$

$\varepsilon(A)$

εA

$\varepsilon r(N)$

(e) (i) In words, define time constant of an RC circuit. (1 mark)

(ii) Find the angular frequency of oscillation in an LC circuit with inductance and capacitance values of 3 mH and 3 μC respectively. (3 marks)

$L = 3\times10^{-3}$

$C = 3\times10^{-6}$

(f) (i) Give one application of a concave mirror? (1 mark)

(ii) A convex lens has radii of curvature of 40 cm is made of glass with refractive index n = 1.65. Find its focal length. (3 marks)

(g) (i) In words, explain the meaning of diffraction of light? (1 mark)

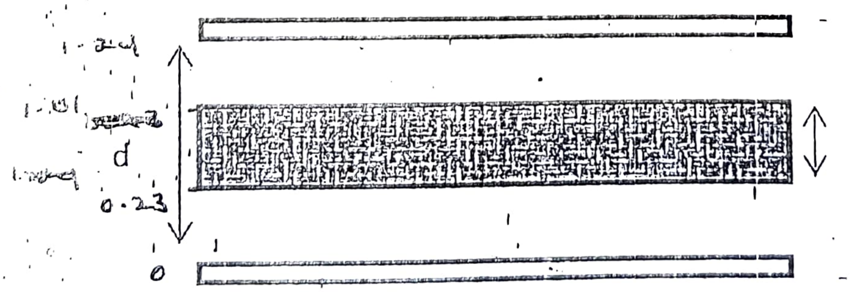
(ii) Coherent light of wavelength 580 nm is incident on a slit of width 0.3 mm. If the observing screen is placed 2 m from the slit, find the position of the first dark fringe. (3 marks)

(h) Differentiate between circularly polarized and elliptically polarized waves. (2 marks)

2. (a) A parallel-plate capacitor of plate area 115 cm^2 and plate separation 1.24 cm. A voltage of 90 V is applied between the plates. The battery is not disconnected and a dielectric slab of thickness 0.78 cm and dielectric constant 2.61 is inserted between the plates as shown below.

$q = \varepsilon_0 E A$

$C = \frac{\varepsilon_0 A}{d}$



$Q = CV$

$C = \frac{Q}{V}$

Fig1

Find

(i) the capacitance before the dielectric was inserted. $C_0 = \frac{\varepsilon_0 A}{d}$ (2 marks)

(ii) the charge on the plates without the dielectric. $q = C_0 V_0$ (2 marks)

(iii) the electric field the air gaps before the dielectric was inserted. (2 marks)

(iv) the electric field in the dielectric after it was inserted. 7253.0645 (2 marks)

(v) the potential difference after the dielectric was inserted. (2 marks)

(vi) the capacitance after the dielectric was inserted. (2 marks)

(vii) the induced charge on the dielectric. (2 marks)

$q - q_1 =$

b) A 2 μF capacitor is charged using a 100 V battery. The battery is disconnected and two uncharged capacitors are connected to it as shown in Fig. 2.

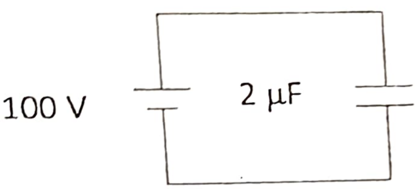


Fig. 2a

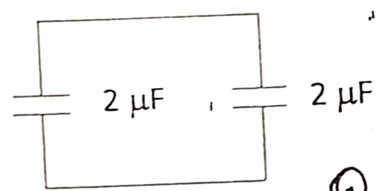


Fig. 2b

$Q = CV$

$Q =$

Find (i) the energy stored in Fig. 1a $U = \frac{1}{2}CV^2$ (3 marks)

(ii) the energy stored in Fig. 1b $\left(U = \frac{q^2}{2C}\right)$ (3 marks)

(a) Consider the circuit in Fig 3 below carrying 2 A. Find the magnetic field at point P.

$B = \frac{\mu_0 I}{4\pi}\left[\frac{x}{\sqrt{x^2+R^2}}\right]_{x^-}^{x^+}$

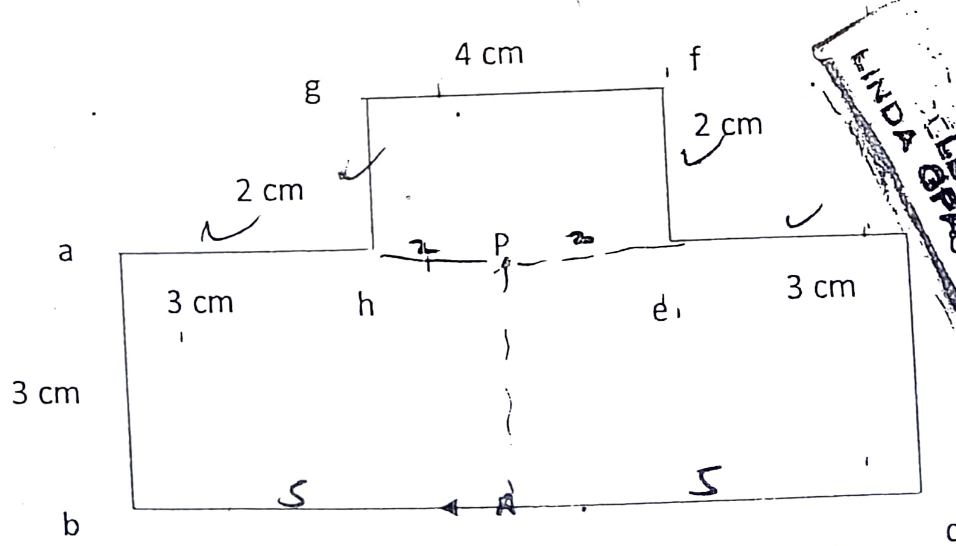


Fig. 3

$q = CV$

$\frac{(CV)^2}{2C}$

$\frac{C^2V^2}{2C}$

$B_{BC} = \frac{\mu_0 I}{}$

(12 marks)

(b) Two conducting rods 0.5 m long roll on conducting rails 0.5 m apart with velocities $v_1 = -12.5\,\hat{y}$ m/s and $v_2 = 8\hat{y}$ m/s. in magnetic field $B = 0.35\hat{z}$ T as shown below

$S = BLV$

$= ILB$

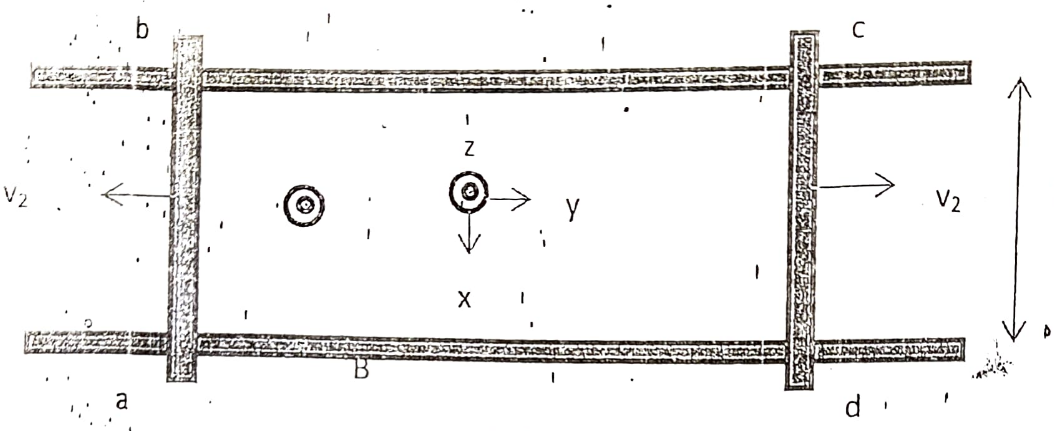


Fig4

Find the voltage of

(i) point b with respect to point a. (3 marks)

(ii) point d with respect to point c. (3 marks)

(iii) point b with respect to point c. (2 marks)

4. (a) Show that the intensity pattern of a single-slit diffraction experiment is given by

$$I_\theta = I_o\left\{\frac{\sin[\pi a\sin(\theta/\lambda)]}{\pi a\sin(\theta/\lambda)}\right\}^2$$

where a is slit width. (14 marks)

(b) A grating has 10000 rulings/inch is illuminated with yellow light at perpendicular incidence. This light contains two closely spaced lines of wavelength 5890.0 and 5895.9 A.

(i) At what angle will the first order maximum occur for the first of these wavelengths? (3 marks)

(ii) What is the angular separation between the first-order maxima for these lines? (3 marks)

SEMESTER II CAT 1. ECU 103 PHYSICS FOR ENGINEERS II.

INSTRUCTIONS: Answer all the three questions.

$\frac{1}{2} \times 9.6 \times 10$ $\frac{1}{2} \times 0.00096 \times 120^{60}$

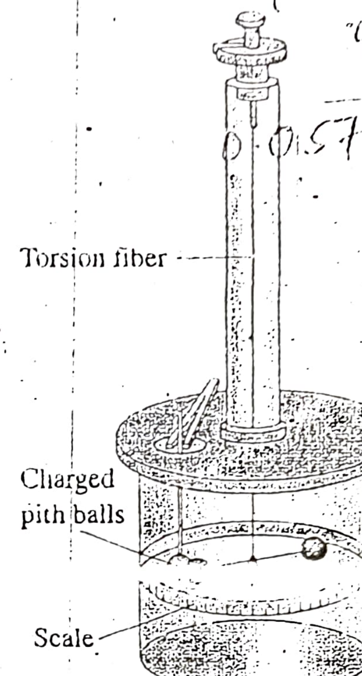
QUESTION 1.

A. Use the sketch of a torsion balance; provided to the right to outline a procedure followed by Charles A. de Coulomb to come up with the famous Coulomb's law $q = \frac{k|q_1q_2|}{r^2}$

$F = \frac{k|q_1q_2|}{r^2}$ $\frac{1}{4\pi\varepsilon_0}$ - permittivity.

$F = \frac{k|q_1q_2|}{r^2}$

$\frac{96}{60}$ 0.576



[5]

B. Give a precise definition of the following electrostatic terms and where applicable, include an equation. $F = \frac{kq}{r^2}\hat{r}$

i. Point Charges, ii). Electric field, iii). Magnetic field, $dB = \frac{\mu_0 I}{2\pi r}$

iv. Dielectric medium, v). Transient Currents

QUESTION 2

A. Capacitor C_1 in the circuit below is charged by source V_0 (not shown on diagram). If $C_1 = 8\mu F$ and $V_0 = 120V$. Calculate the;

i. charge Q_0 on the capacitor plates, ii). energy of the capacitor [4]

B. After the switch S is closed, both capacitors get charged to Q_1 and Q_2 respectively such that $Q_1 + Q_2 = Q_0$

i. Calculate the final potential difference (V) between the plates of the two capacitors [3]

ii. What would be the values of Q_1 and Q_2 respectively? [2]

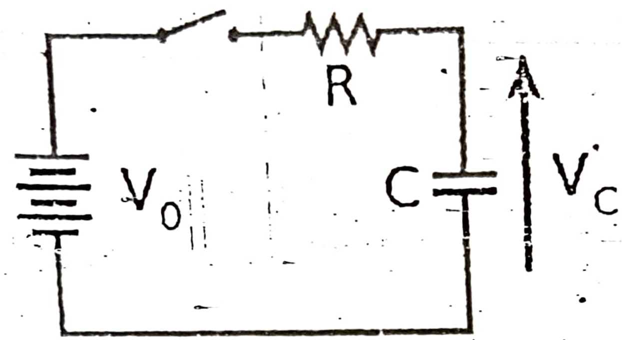
iii. Calculate the final energy of the combined circuit. [4]

iv. Compare and comment on the energy value obtained here with the one of Aii. [2]

QUESTION 3.

Answer the questions that follow, by considering the circuit diagram below containing the components shown.

A. What general name is given to this type of circuit and why? [1]



B. If the capacitor is initially uncharged while the switch is open, and the switch is closed at time t_0, write an expression for the value of V_0 in terms of R and C. [illegible]

$$V_0 = V_R(t) + V_C(t)$$

C. From the expression obtained in part (B) above, how do we get a first-order differential equation of the type written below? [illegible]

$$RC\frac{di(t)}{dt} + i(t) = 0$$

D. What special name is given to the expression RC for the above circuit? [1]

UNIVERSITY EXAMINATIONS 2014/2015

SECOND SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE (ELECTRICAL AND ELECTRONIC ENGINEERING)

ECU 103: PHYSICS FOR ENGINEERS II

DATE: Thursday, 16th April 2015 **TIME: 11.00 a.m. – 1.00 p.m.**

INSTRUCTIONS:

1. Answer questions ONE and any other TWO questions.
2. The following constants and table may be useful:

$c = 3 \times 10^8$ m/s, $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m, $\mu_0 = 4\pi \times 10^{-7}$ H/m

Q1.(a) (i) In words, explain the meaning of superposition principle in electrostatics? (1 mark)

(ii) Three identical charges are placed on the vertices of an equilateral triangle. If two charges exert a force of 0.001 N each on the third. What is the resultant force on the third charge? (3 marks)

(b) (i) In words, explain the meaning of Biot-Savart law? (1 mark)

(ii) A curved wire subtending an angle $\pi/4$ at the center of curvature of radius 5 cm, is carrying 5 A. Find the field at the centre of curvature. (3 marks)

(c) (i) In words, explain the meaning of Gauss's law? (1 mark)

(ii) Use Gauss's law to derive the expression for capacitance per unit length of a cylindrical capacitor. (3 marks)

(d) (i) In words, explain the meaning of Amperes law? (1 mark)

(ii) A wire of radius 3 mm carrying 2 A. Find the magnetic field at a point 2 mm from its center? (3 marks)

(e) (i) In words, explain the meaning of a filter? (1 mark)

ii) Show that an RC circuit can act as a low pass filter. (3 marks)

(f) (i) Give one application of a convex mirror? (1 mark)

(ii) A convex lens has radii of curvature of 40 cm is made of glass with $n = 1.65$. Find its focal length. (3 marks)

(g) (i) In words, explain the meaning of Huygen's principle? (1 mark)

(ii) A light beam passes from medium 1, with an index of refraction n_1, through a thick slab of material whose index of refraction is n_2, and emerges from the slab into medium 1. Show that the emerging beam is parallel to the incident beam. (3 marks)

(h) Differentiate between linearly polarized and circularly polarized waves. (2 marks)

Q2. (a) A parallel plate capacitor of area 115 cm^2 and plate separation 1.24 cm is filled with removable dielectric slab with $\varepsilon_r = 2.6$. The capacitor is given a charge Q with the slab removed and not disconnected from the battery of $V_\sigma = 90$ V. Find

(i) the charge on the plates without the slab. (2 marks)

(ii) the charge on the plates with the slab inserted. (2 marks)

(iii) the field without the slab. (2 marks)

(iv) the field with the slab inserted. (2 marks)

(v) induced charge with the slab inserted (2 marks)

(b) A 2 µF capacitor is charged using a 100 V battery. The battery is disconnected and two uncharged capacitors are connected to it as shown in Fig. 1.

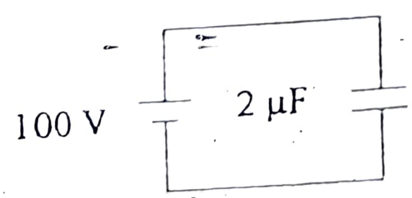


Fig. 1a

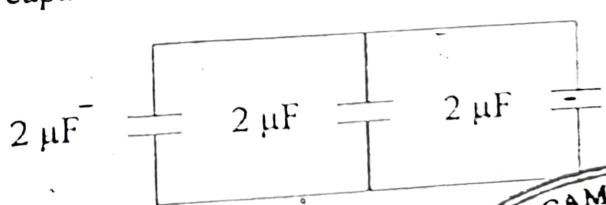


Fig. 1b